



The Wrong Direction

A move goes the wrong way — fix it

Name: _____

Date: _____

★ I can find a move that goes the wrong way and fix its direction.

The goal: both reach 2

Bit (green) and Pixel (purple) both start on the green flag. They should BOTH reach number 2. One script is correct, but the other has a move going the WRONG WAY — forward when it should go back. Read both, find the wrong direction, and fix it.

Number path 0–5



Bit's script (correct)

🚩 when green flag clicked

forward 2

Pixel's script (has a bug)

🚩 when green flag clicked

forward 2

1 Run both

Where does each land? Bit (home 0, forward 2): ____ Pixel (home 4, forward 2): _____. (Remember: a sprite cannot go past 5.) Which one misses the goal 2?

2 Find the wrong direction

Pixel starts at 4 and needs to reach 2. Is 2 in FRONT of Pixel or BEHIND it? So which way should Pixel move — forward or back?

3 Fix it

Write Pixel's fixed move: ____ 2 (forward or back?). After the fix, where does Pixel land? Do both reach the goal now?

Big idea

A move can have the right number but the wrong DIRECTION. Look at where the sprite starts and where the goal is: if the goal is behind it, the move must go back, not forward. Here Pixel needed back 2, not forward 2.



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Quick facilitation notes & answer key

What this worksheet teaches

Students fix a wrong-direction bug. The goal is for both to reach 2; Bit is right, but Pixel moves forward (away from 2) when it should move back. Students reason about which way the goal is. No coding experience needed.

Before you start (no computer needed)

No computer needed — paper and a pencil. You read the prompts aloud. Optional: two counters on a number line 0 to 5 (green Bit, purple Pixel) so children can trace both sets of moves and see which one misses the goal.

How to lead it (about 20 minutes)

1. Show them (you do). Is the goal in front of Pixel or behind it? That tells you forward or back.
2. Do one together. Exercise 1 — Bit reaches 2; Pixel goes forward to 5 (overshoots).
3. Let them try. Students decide the direction and write the fix (Ex 2–3).

Answer key (these are the verified correct answers)

Exercise 1 — Bit: 0, forward 2 → 2 (reaches goal). Pixel: 4, forward 2 → 5 (the path stops at 5; Pixel overshoots and misses 2).

Exercise 2 — the goal 2 is behind Pixel (Pixel is at 4), so Pixel should move back, not forward.

Exercise 3 — fixed move back 2 → Pixel lands on 2. Both reach 2. The number (2) was fine; only the direction changes.

If students get stuck — and ways to adjust

Watch for: changing the number instead of the direction. Here only the direction is wrong.

Make it easier: stand Pixel's counter on 4 and point toward the goal to find the direction.

Make it harder: ask what would happen if both the number and direction were wrong.

Ontario curriculum link (official wording)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential and concurrent events.

In plain terms: students write and run instructions, including two sets that run at the same time.

C3.2 — read and alter existing code, including code that involves sequential and concurrent events, and describe how changes to the code affect the outcomes.

In plain terms: students read two sets of instructions, find and fix the wrong one, and explain the fix.

You'll know it worked when

The child flips Pixel's direction so it reaches the goal.